

Project Proposal

femora

AI Powered Women's Health & Wellness Application

By

Arshia Naseer – 232428

Ali Haider Bilal – 232398

Ammad Sajjad – 232497

Supervisor

Mustabshera Fatima



Department of Computer Science
Air University Islamabad (AU)

Table of Contents

Supervisor Meeting Log:	3
0. Updated Section	5
STREAMLINED MODULES	5
Domain 1: Menstrual Cycle	5
Domain 2: PCOS Detection	5
Domain 3: Breast Cancer	5
1. Introduction	6
2. Related System Analysis / Literature Review	6
3. Problem Statement	7
4. Problem Solution / Objectives of the Proposed System	8
4.1 Objectives	8
5. Scope	8
6. Modules	9
6.1 User Onboarding and Health	9
6.2 Menstrual Cycle Tracking and Prediction	9
6.3 Breast Cancer Awareness and Detection	9
6.4 PCOS / PCOD Risk Detection	10
6.5 Pregnancy Care	10
6.6 Harmonal Health Insights	10
6.7 AI Healthcare Companion	10
6.8 Notifications and Reminders	10
6.9 Reports and Health Insights Dashboard	11
6.10 Daily Symptom and Mood Tracker	11
7. System Limitations/Constraints	11
8. Data Gathering Approach	11
9. Tools and Technologies	12
10. Project Stakeholders and Roles	12
11. Module based Work Division	14
12. Gantt Chart	14
14. Flow-Chart	15
14. Mockups	16
14.1 Home / Dashboard	16

14.2 Cycle Calendar + Daily Tracker	17
14.3 PCOS Risks + Hormonal Insights	18
14.4 Breast Cancer Module	19
14.5 AI Health Companion	20
14. References	21

Supervisor Meeting Log:

Meeting / Date	Discussion	Task Assignment	Supervisor Signature
Meeting 01 Date: 11 – 05 – 2026	Discussed FYP ideas, finalized Femora as project topic. Evaluated feasibility, constraints, and target users.	Write initial proposal. Research similar apps (Flo, Clue).	
Meeting 02 Date: 12 – 05 – 2026	Reviewed research papers on menstrual cycle prediction, PCOS detection, breast cancer CNNs, and AI in women's healthcare.	Finalize modules based on literature. Compile related system analysis.	
Meeting 03 Date: 15 – 05 – 2026	Explored datasets (Fehring, IEEE, BUSI, WISCOSINE). Identified limitations: small size, demographic bias, missing fields. Discussed SMOTE.	Download and analyze datasets. Document complications and solutions.	
Meeting 04 Date: 17 – 05 – 2026	Finalized AI models: Random Forest + LSTM for cycle prediction, XGBoost for PCOS, ResNet50 for breast cancer. Discussed train/test split.	Finalize model selection. Prepare data preprocessing pipeline.	
Meeting 05 Date: 18 – 05 – 2026	Reviewed and finalized scope document covering all modules, tools, work division, and Gantt chart. Reviewed presentation slides.	Submit scope document. Complete FYP presentation with supervisor feedback.	
Meeting 06 Date: 29 – 06 – 2026	Trimming some modules of each domain meeting requirements of the panel	Submitted new scope document with updated section	

0. Updated Section

STREAMLINED MODULES

Domain 1: Menstrual Cycle

- Log cycle dates & symptoms
- General Model: Random Forest (70% accuracy day 1)
- Personalized Model: LSTM after 3+ cycles (85-90%)

Domain 2: PCOS Detection

- Risk questionnaire: cycle irregularity, BMI, symptoms
- XGBoost classifier → Low/Medium/High risk score

Domain 3: Breast Cancer

- Module A: Ultrasound upload → ResNet50 CNN classification
- Module B: Symptom questionnaire → XGBoost risk score

DATASET USAGE & PRIVACY

Dataset	Size	Used For	Verified Source	Privacy
Menstrual Cycle 1. Fehring Dataset 2. IEEE Menstrual Cycle Dataset mcPHASES	86 women × 12 cycles = 1,032 rows	Cycle length prediction (General + Personalized model)	Fehring, R. J. et al. (2013). Marquette University e-Publications. [5]	PII stripped
	42 women, 2,937 daily rows	Daily phase and symptom detection model (LSTM)	IEEE Dataport (2022) [10]	
PCOS Dataset	541 labeled women	PCOS risk classification model	Kaggle — Real hospital records, Kerala, India. [12]	Encrypted locally
Breast Cancer 1. BUSI Dataset 2. Breast-Lesions-USG 3. WISCONSIN Dataset	780 images, 600 women	Breast cancer CNN classifier	Al-Dhabyani et al. (2020), Data in Brief [24]	Deleted after processing
	256 biopsy-confirmed images	Breast cancer CNN classifier (combined with BUSI)	Pawłowska et al. (2024), Cancer Imaging Archive [26]	
	569 patient records, 30 features	Breast cancer symptom risk assessment (XGBoost)	Wolberg et al. (1993). UCI Machine Learning Repository. [22]	

1. Introduction

Women's health remains significantly underserved in the digital healthcare space. Research by Epstein D. A. (2017) [1] found that while menstrual cycle tracking apps are widely used, they are most accurate only for women with very regular cycles, whereas the women who need these applications most are those with irregular cycles. A study examining 10 menstrual cycle tracker apps by Worsfold et al. (2021) [2] found that 67% of ovulation predictions were 2-9 days early, demonstrating fundamental inaccuracy in current prediction approaches.

Three major health conditions further demonstrate the need for Femora. First, PCOS affects approximately 9.2% of women globally [4], with South Asian women at disproportionately higher risk, yet no widely accessible AI-powered screening tool exists for this population. Second, breast cancer

is the most commonly occurring cancer in women worldwide [21]. In 2022 alone, 2.3 million women were diagnosed with 670,000 deaths [16]. In Pakistan specifically, breast cancer peaks at ages 40–50, a decade earlier than in Western populations [17], making early awareness tools critically important. Third, Pregnancy complications such as pre-eclampsia and eclampsia accounts for approximately one-third of maternal deaths in Pakistani tertiary hospitals [18] conditions that benefit significantly from early mobile-based monitoring. No single existing platform addresses all three of these needs together.

Femora aims to address this gap by providing an AI-powered women's health platform integrating menstrual cycle prediction using adaptive personalization, PCOS risk detection, pregnancy care, and a breast cancer module combining CNN-based ultrasound image classification with a symptom-based risk assessment model, all in one accessible application requiring no hardware.

2. Related System Analysis / Literature Review

Application Name	Weakness	Proposed Project Solution
Flo Health	Zhaunova et al. (2023) [6] found Flo relies on past cycle averages with no lifestyle integration. A 2025 GDPR audit [7] also reported that Flo uses statistical averaging without accounting for stress, sleep, or BMI. No breast cancer awareness. No PCOS detection.	Femora uses Random Forest ML trained on the Fehring peer-reviewed clinical dataset [8] incorporating stress, sleep, and BMI. Includes PCOS risk scoring, breast cancer image classification [24], and pregnancy care in one unified platform.
Clue	Oxford Open Digital Health (2025)	Femora integrates lifestyle-aware cycle

	[3] reported that Clue does not account for lifestyle factors in cycle predictions. A systematic review [19] highlighted the lack of breast cancer awareness features in menstrual health applications.	prediction, PCOS risk scores, and a breast cancer module combining CNN ultrasound classification [24] with symptom-based risk assessment [25] — unavailable in any existing app.
Natural Cycles	A 2025 ScienceDirect ML study [9] reported that Natural Cycles requires daily basal body temperature (BBT) monitoring, making it less practical for users with irregular cycles or PCOS. The platform mainly focuses on fertility tracking only.	Femora operates without requiring external hardware and provides integrated menstrual tracking, PCOS risk assessment, pregnancy care, and breast cancer awareness features.
General Period Apps	Worsfold et al. (2021) [2] tested 10 apps, 67% of ovulation predictions were inaccurate by 2-9 days. Epstein et al. (2017) [1] also reported that most apps perform poorly for women with irregular cycles.	Femora uses adaptive personalization to improve prediction accuracy over time [11] while combining cycle tracking, hormonal health insights, PCOS detection, pregnancy care, and breast cancer awareness in one system.

3. Problem Statement

Existing women’s health applications mainly focus on basic menstrual tracking using generalized statistical averages. These applications fail to provide accurate predictions for women with irregular cycles [1] because they do not adapt to individual lifestyle factors such as stress, sleep, BMI, and symptom variations [3]. As a result, many users receive inaccurate ovulation and cycle predictions, reducing the reliability and usefulness of current menstrual cycle tracking systems.

PCOS is one of the most common hormonal disorders affecting women, particularly in South Asian populations. Despite its prevalence, diagnosis is often delayed because accessible early screening tools are limited. Current health applications do not analyze cycle irregularity, symptom history, or BMI to provide intelligent PCOS risk assessment.

Breast cancer awareness and early detection support remain limited in women’s health applications. Most menstrual tracking systems do not include breast health monitoring, symptom-based risk assessment, or medical image analysis capabilities such as ultrasound classification. As a result, users must rely on separate systems for reproductive health and breast health management. This separation limits early awareness and prevents users from accessing multiple women’s health services in one platform.

Pregnancy care applications mainly provide static guidance and reminders but rarely include intelligent symptom monitoring or anomaly detection. Additionally, pregnancy monitoring systems

are usually separated from menstrual and hormonal health platforms, forcing users to switch between multiple applications during different stages of health management.

4. Problem Solution / Objectives of the Proposed System

Femora proposes an AI-powered mobile application using machine learning trained on real peer-reviewed clinical datasets. Li et al. (2022) [\[11\]](#) (Journal of the American Medical Informatics Association) demonstrated that predictive models accounting for user adherence significantly outperform simple averaging.

4.1 Objectives

- To develop an AI-powered menstrual cycle tracking system capable of predicting cycle length, period duration, ovulation window, and daily cycle phases using clinical and symptom-based data
- To implement adaptive personalization that improves prediction accuracy over time based on individual user history and symptom patterns.
- To develop a PCOS/PCOD risk assessment module using factors such as cycle irregularity, BMI, and user-reported symptoms to provide early risk awareness.
- To implement a breast cancer awareness module that supports AI-based ultrasound image classification and symptom-based risk assessment for early awareness assistance.
- To include a pregnancy care module that offers trimester-based guidance, symptom monitoring, reminders, and anomaly flagging for high-risk conditions.
- To provide hormonal health insights by analyzing symptom patterns across different menstrual cycle phases.
- To develop an AI healthcare companion capable of answering user health-related queries and providing personalized wellness support.
- To provide visual health reports and dashboards showing cycle trends, symptom history, and prediction analytics
- To ensure secure handling, privacy, and ethical management of sensitive user health data.

5. Scope

The scope of Femora includes the development of an AI-powered mobile healthcare application that provides menstrual cycle prediction, PCOS risk assessment, hormonal health insights, pregnancy care, breast cancer awareness and detection support, and an AI healthcare companion within a single platform. The system utilizes machine learning models trained on peer-reviewed clinical datasets for cycle prediction, daily phase detection, breast cancer image classification, and symptom-based risk assessment to deliver personalized healthcare insights and

early risk awareness. Predictions and recommendations are generated using user-logged symptoms, health profiles, and optionally uploaded medical images, while the system operates without requiring external hardware. Wearable device integration and Pakistan-specific dataset expansion are considered future enhancements.

6. Modules

6.1 User Onboarding and Health

- Collect baseline health information including age, BMI, cycle regularity, and existing conditions such as PCOS.
- Build personalized user health profiles to initialize AI prediction models and address the cold-start problem.
- Secure account creation with email authentication and profile management.

6.2 Menstrual Cycle Tracking and Prediction

- Allow users to log menstrual cycle details including Menstruation dates, cramps, and daily symptoms, addressing tracking needs identified by Epstein et al. (2017) [1].
- General Model predicts next cycle length and period duration from day one. 70% accuracy on day one, improving with logged cycles (Random Forest, trained on Fehring dataset [8]).
- Personalized Model adapts to specific woman's cycle pattern after 3+ cycles, giving 85-90% accuracy, (LSTM, trained on individual user history).
- Display menstrual phases and predictions through an interactive calendar interface.
- Detect delayed or irregular cycles and provide lifestyle-based explanatory insights, addressing the documented gap in Oxford Open Digital Health (2025) [3].

6.3 Breast Cancer Awareness and Detection

The breast cancer module consists of three integrated components:

- Component A CNN Image Classification: User uploads a hospital ultrasound image (PNG/JPG). A ResNet50 CNN model using transfer learning, fine-tuned on combined BUSI dataset [24] and Breast-Lesions-USG dataset [26]
- Component B Symptom-Based Risk Assessment: For users without an ultrasound scan, a questionnaire-based XGBoost model trained on the Wisconsin dataset [25] computes a risk score based on age, family history, menstrual history, and self-reported symptoms.

- Component C AI Chatbot: Results from Component A or B feed into an NLP-based AI chatbot that explains findings in simple language and answers follow-up questions.
- Provide monthly breast self-examination reminders
- Generate awareness and risk assessment only and not medical diagnosis [23]

6.4 PCOS / PCOD Risk Detection

- Compute PCOS risk scores using cycle irregularity, BMI, and self-reported symptoms including acne, weight gain, and abnormal hair growth.
- Classify risk as Low (0–30%), Medium (30–60%), or High (60–100%) with educational guidance and professional consultation recommendation.
- Provide risk awareness only and not clinical diagnosis, considering the high global prevalence of PCOS reported in recent studies [4]

6.5 Pregnancy Care

- Switch to pregnancy mode when user marks confirmed pregnancy. Validated by evidence that mobile-based pregnancy monitoring is feasible and effective in low-resource settings including Pakistan [18].
- Track pregnancy week, trimester progress, and related symptoms.
- Allow daily logging of pregnancy symptoms such as nausea, swelling, fetal movement, blood pressure changes. Pre-eclampsia risk monitoring directly informed by Feroz et al. (2022) [18].
- Flag critical symptoms requiring immediate medical attention

6.6 Hormonal Health Insights

- Analyze daily logged symptoms across cycle phases.
- Provide phase-specific hormonal health insights and wellness recommendations.
- Flag persistent hormonal imbalance symptoms for medical consultation.

6.7 AI Healthcare Companion

- Provide personalized health reminders including medication, hydration, and self-examination notifications.
- Answer health-related queries using an NLP-based conversational interface through Claude/GPT APIs.
- Offer emotional wellness support during menstrual and pregnancy phases.

6.8 Notifications and Reminders

- Send reminders for upcoming menstruation, ovulation windows, and fertility phases.
- Provide daily symptom logging and medication reminders.
- Send monthly breast self-examination and pregnancy care reminders where applicable.

6.9 Reports and Health Insights Dashboard

- Display cycle history, symptom patterns, hormonal trends, and prediction analytics.
- Generate downloadable health reports for healthcare consultation and personal monitoring.
- Provide visual dashboards for improved health tracking and analysis.

6.10 Daily Symptom and Mood Tracker

- Allow users to log daily symptoms, mood changes, sleep patterns, stress levels, and related health indicators.
- Use logged data to improve adaptive personalization and AI prediction accuracy.
- Display symptom trends and historical analysis using visual charts and reports.

7. System Limitations/Constraints

- Femora does not provide clinical diagnoses. All risk scores and image classifications are for awareness purposes only and must be followed up with qualified medical professionals.
- Prediction accuracy on day one is approximately 70%, improving to 90% after 3+ cycles. As noted by Worsfold et al. (2021) [2], there are currently no universally agreed-upon criteria for evaluating period prediction app accuracy. Femora's metrics are based on held-out real patient test data.
- PCOS and breast cancer risk scores from the symptom-based model are derived from self-reported data only they do not incorporate blood tests, 3d-scans or clinical examination data.
- Wearable device integration for real-time physiological metrics is outside the current project scope and is planned as a future enhancement.
- The system relies on consistent and honest user input. Inaccurate or irregular logging will reduce prediction quality.

8. Data Gathering Approach

Dataset	Size	Used For	Verified Source
Fehring Dataset	86 women $\times$ 12 cycles = 1,032 rows	Cycle length prediction (General)	Fehring, R. J. et al. (2013). Marquette University e-Publications. [5]

		+ Personalized model)	
IEEE Menstrual Cycle Dataset mcPHASES	42 women, 2,937 daily rows	Daily phase and symptom detection model (LSTM)	IEEE Dataport (2022) [10]
PCOS Dataset	541 labeled women	PCOS risk classification model	Kaggle — Real hospital records, Kerala, India. [12]
BUSI Dataset	780 images, 600 women	Breast cancer CNN classifier	Al-Dhabyani et al. (2020), Data in Brief [24]
Breast-Lesions-USG	256 biopsy-confirmed images	Breast cancer CNN classifier (combined with BUSI)	Pawłowska et al. (2024), Cancer Imaging Archive [26]
WISCONSIN Dataset	569 patient records, 30 features	Breast cancer risk assessment (XGBoost)	Wolberg et al. (1993). UCI Machine Learning Repository. [22]

9. Tools and Technologies

Category	Tools / Technology	Rationale
Frontend	Flutter /React Native	Cross-platform mobile development for iOS and Android from a single codebase.
Backend	Node.js, Microservices	Scalable, event-driven backend architecture suitable for health data and image upload APIs.
AI / ML (Tabular)	Python, Scikit-learn, XGBoost	Random Forest for cycle prediction; XGBoost for PCOS and breast cancer risk assessment.
AI / ML (Image)	Python, TensorFlow, LSTM	LSTM for personalized cycle prediction and daily phase detection.
AI Companion	Python, TensorFlow, Keras, OpenCV, ResNet50	ResNet50 transfer learning for CNN breast cancer ultrasound image classification (BUSI + Breast-Lesions-USG datasets).
Database	Claude / GPT API, NLP	NLP-based conversational AI for healthcare companion and breast cancer result explanation.
Cloud	PostgreSQL / Firebase	Secure structured storage for health records and uploaded images with real-time sync.
Data Processing	AWS / Google Cloud	Scalable model hosting, image storage, and

		API deployment.
--	--	-----------------

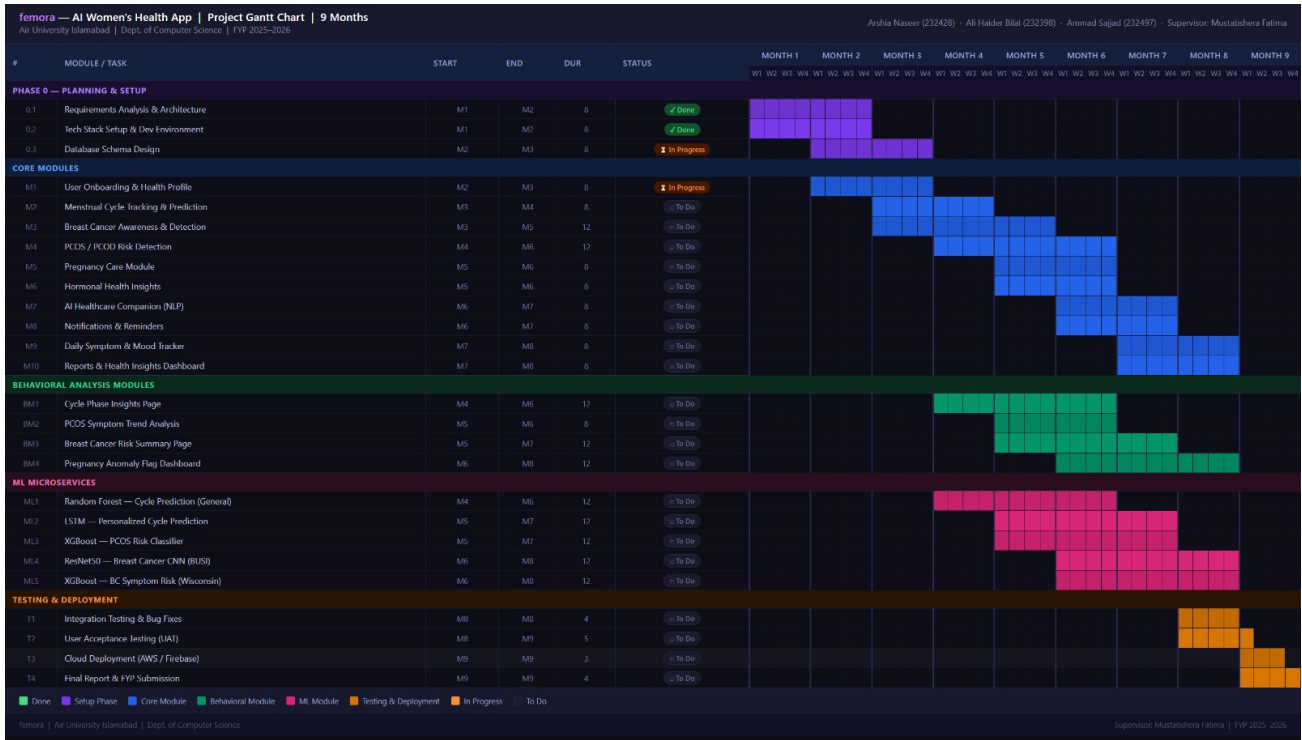
10. Project Stakeholders and Roles

Project Sponsor	Air University Islamabad
Supervisor	Mustabshera Fatima
Development Team	• Arshia Naseer • Ali Haider Bilal • Ammad Sajjad

11. Module based Work Division

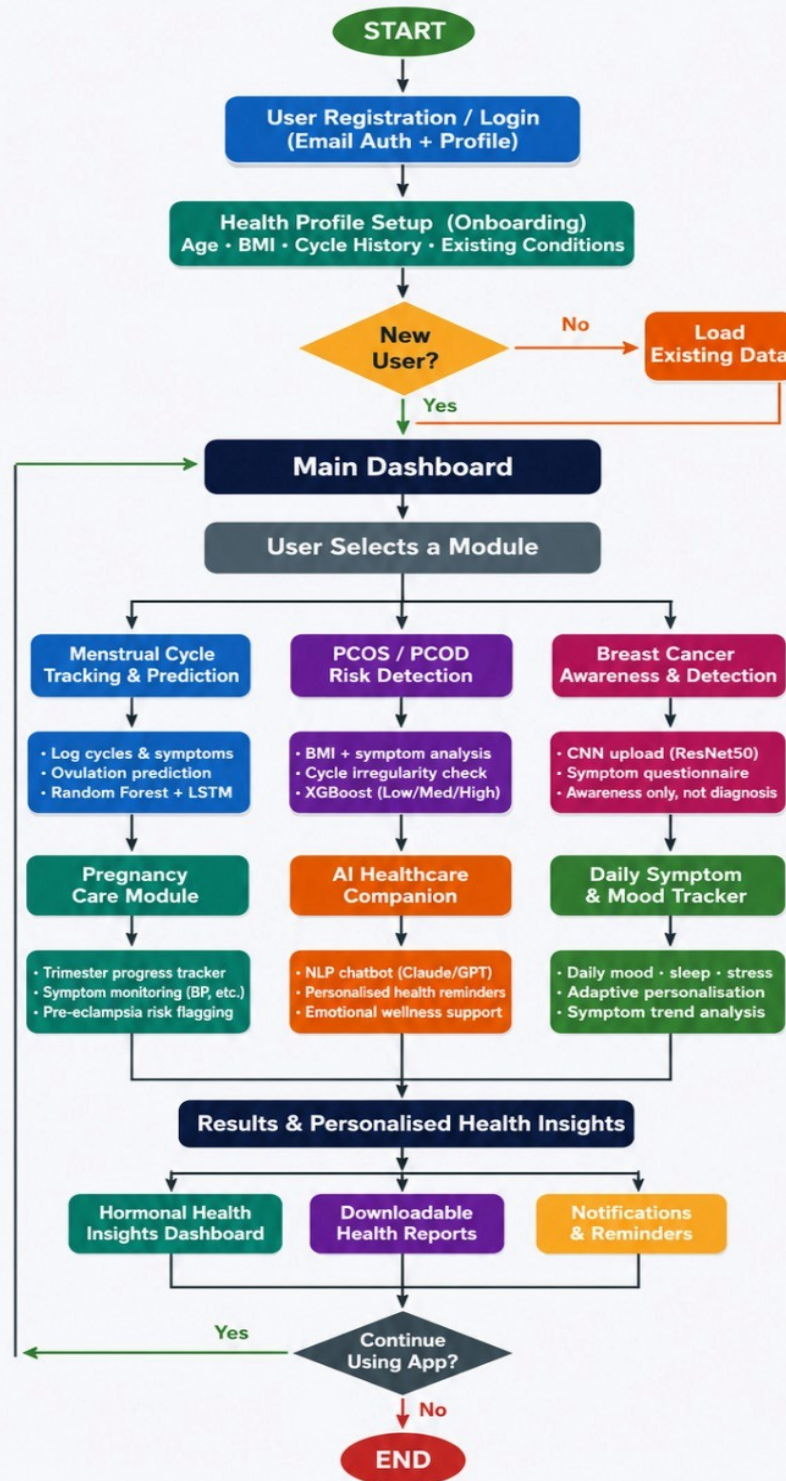
Student Name	Student Registration Number	Responsibility/ Feature
Arshia Naseer	232428	Backend Lead Node.js microservices, database design, user authentication, AI companion chatbot (NLP API), pregnancy care module, breast cancer CNN model deployment, Wisconsin risk assessment model
Ali Haider Bilal	232398	AI/ML Lead Menstrual cycle prediction models (General + Personalized), PCOS risk model, SMOTE pipeline, data cleaning, model training and evaluation, backend API integration.
Ammad Sajjad	232497	Frontend Lead Mobile UI/UX design, cycle calendar, daily tracking screens, breast cancer image upload UI, health dashboard, notifications, user onboarding. Responsive Design Implementation

12. Gantt Chart



14. Flow-Chart

Femora — AI Women's Health App | System Flow Diagram



Module Colour Key

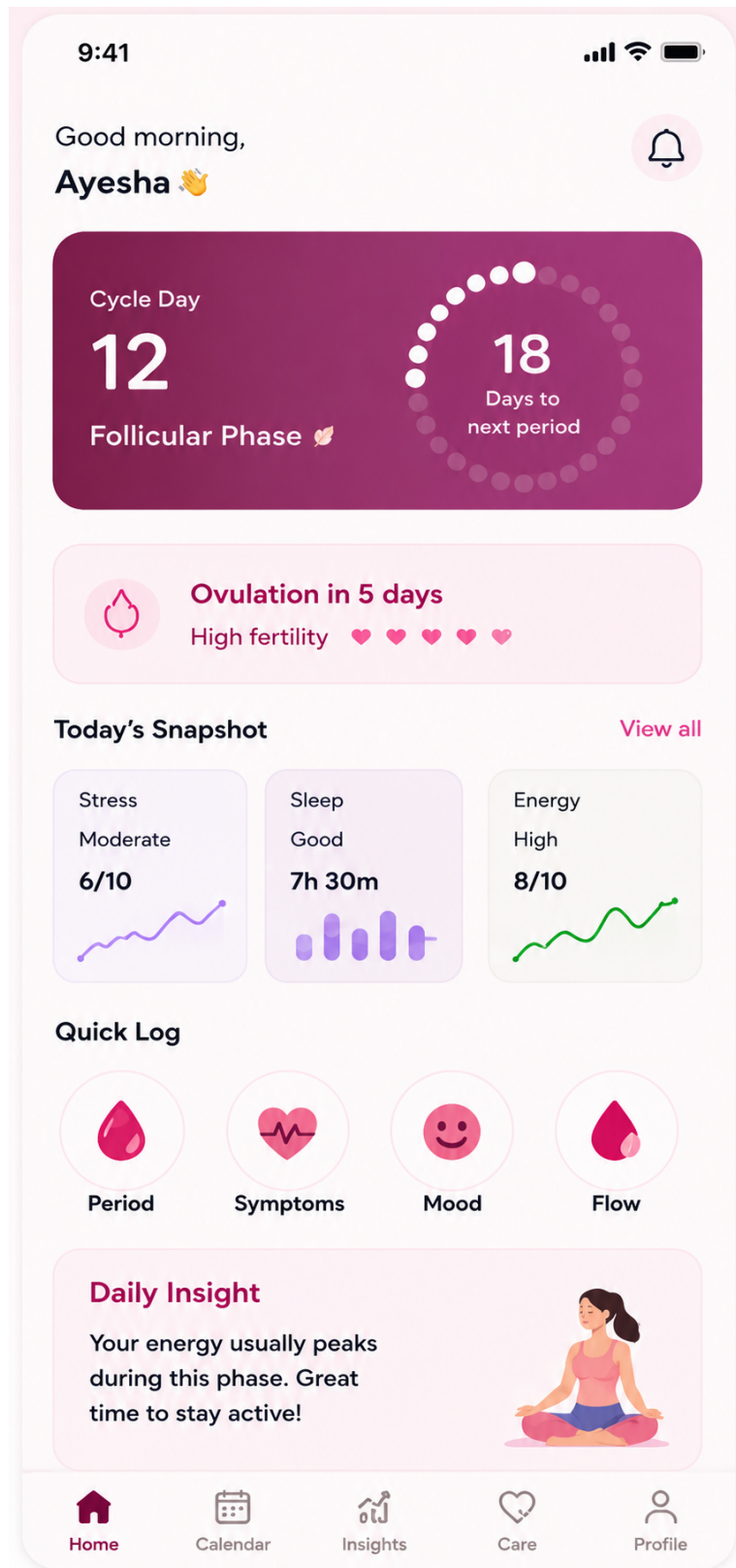
■ Menstrual Cycles
 ■ PCOS
 ■ Breast Cancer
 ■ Pregnancy Care
 ■ AI Companion
 ■ Symptom Tracker
 ■ Notifications
 ■ Decision Flow

Arshia Naseer (232428) • Ali Haider Bilal (232398) • Ammad Sajjad (232497)

Supervisor: Mustabshera Fatima | Air University Islamabad | FYP 2025–2026

14. Mockups

14.1 Home / Dashboard



Description:

14.2 Cycle Calendar + Daily Tracker

9:41

←

Cycle Calendar

⚙️

<

May 2025

>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
27	28	29	30	1	2	3
4	5	6	7	8 ...	9 ...	10
11	12	13	14	15	16	17 .
18	19	20	21	22	23	24
25	26	27	28	29	30	31

● Period

● Follicular

● Ovulation

● Luteal

○ Today

Today • Cycle Day 12

Symptoms

Flow

Mood

Others

Cramps

Bloating

Headache

Fatigue

Acne

Backache

Tender Breasts

Nausea

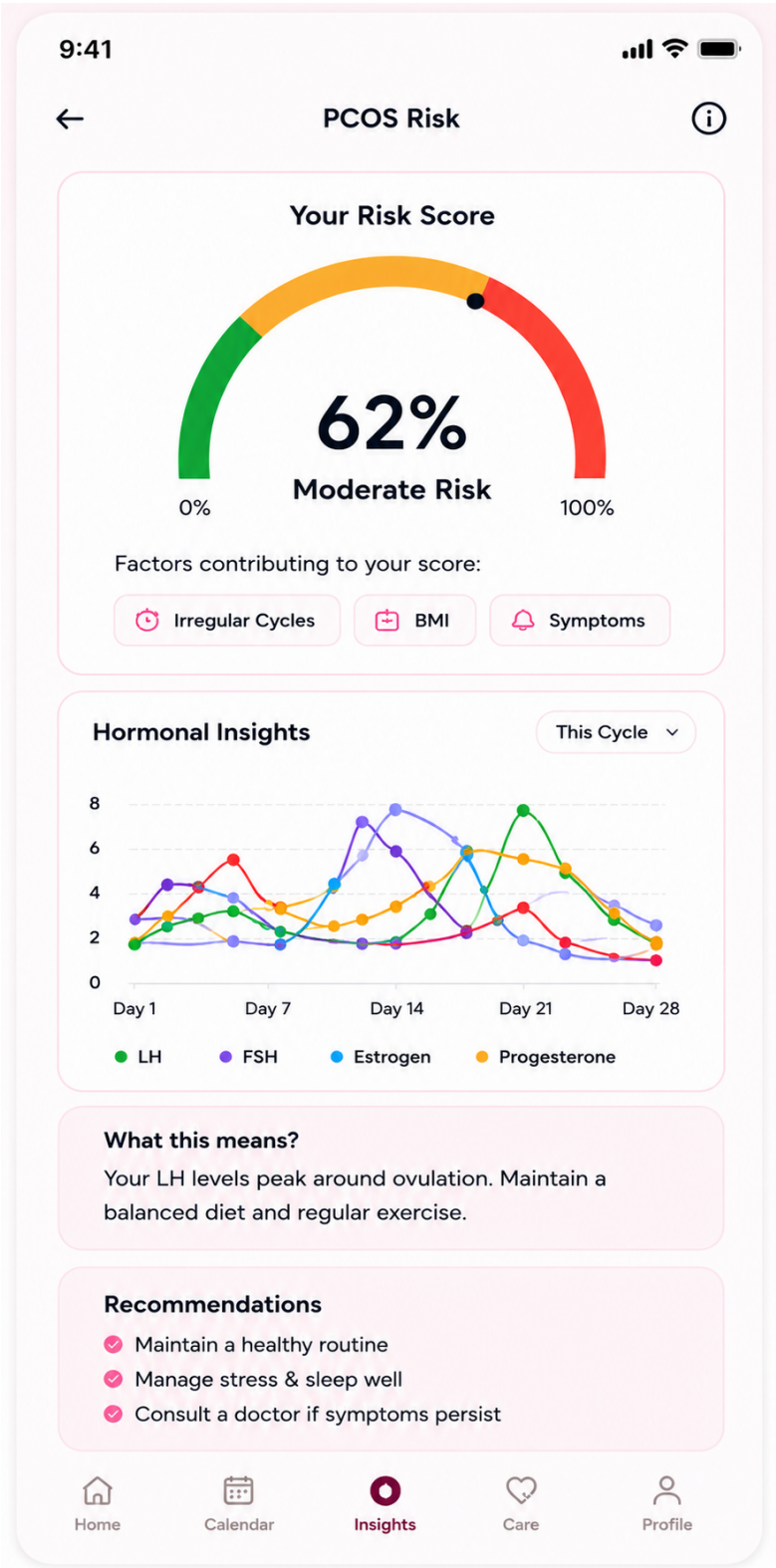
Notes

How are you feeling today?

Save Today's Log

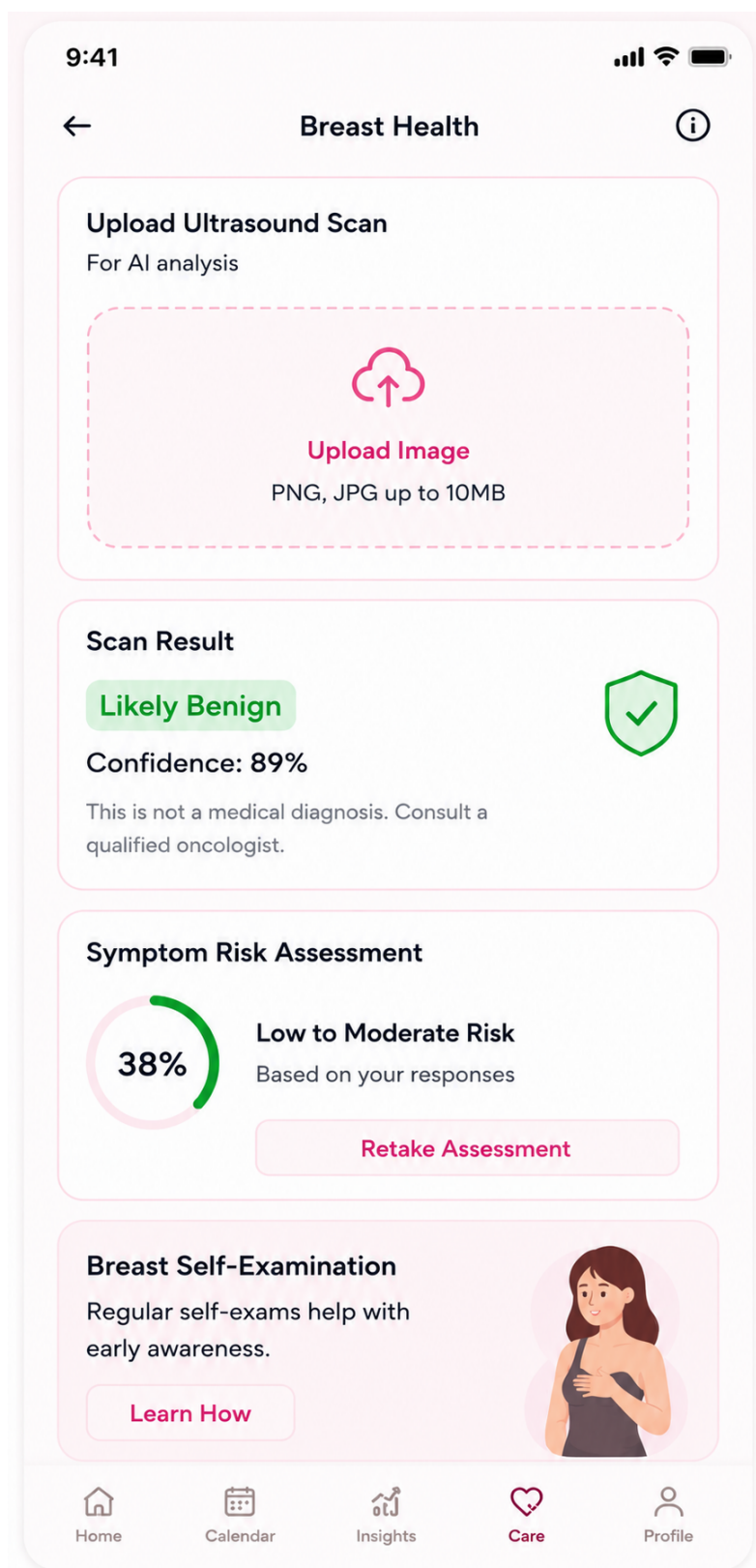
Description:

14.3 PCOS Risks + Harmonal Insights



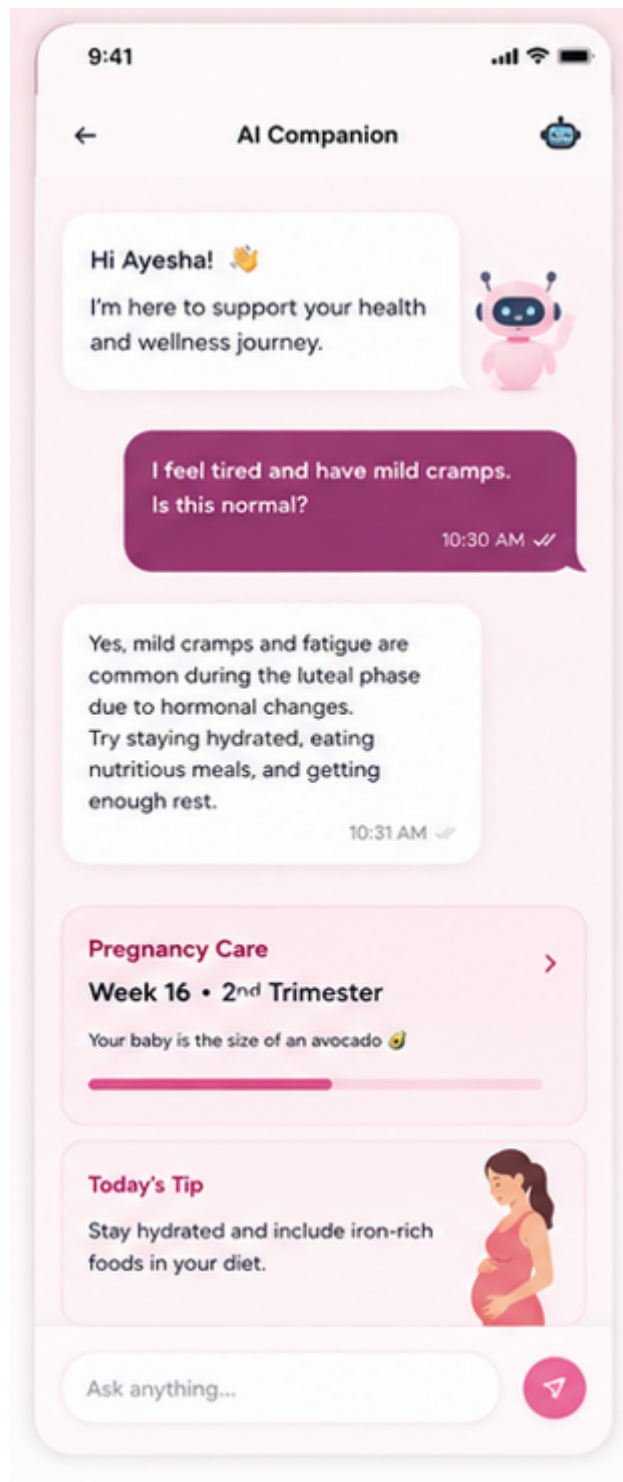
Description:

14.4 Breast Cancer Module



Description:

14.5 AI Health Companion



Description:

14. References

- [1] Epstein, D. A., Lee, N. B., Kang, J. H., et al. (2017). Examining Menstrual Tracking to Inform the Design of Personal Informatics Tools. Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5432133/>
- [2] Worsfold, L., Marriott, L., Johnson, S., & Harper, J. C. (2021). Period tracker applications: What menstrual cycle information are they giving women? Women's Health. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8504278/>
- [3] Flowing data: women's views and experiences on privacy and data security when using menstrual cycle tracking apps. Oxford Open Digital Health (2025). <https://academic.oup.com/oodh/article/doi/10.1093/oodh/oqaf011/8134275>
- [4] Global prevalence of polycystic ovary syndrome in women worldwide: a comprehensive systematic review and meta-analysis (2024). <https://www.researchgate.net/publication/381732781>
- [5] Fehring, Richard J., "Menstrual Cycle Data" (2012). *Randomized Comparison of Two Internet-Supported Methods of Natural Family Planning*. 7. https://epublications.marquette.edu/data_nfp/7
- [6] Zhaunova, L., Bamford, R., Radovic, T., et al. (2023). Characterization of Self-reported Improvements in Knowledge and Health Among Users of Flo Period Tracking App. JMIR mHealth and uHealth, 11, e40427. <https://doi.org/10.2196/40427>
- [7] Experiential case study audit of three popular period trackers using GDPR and intimate privacy assessment criteria. ScienceDirect (2025). <https://www.sciencedirect.com/science/article/pii/S0010782425004263>
- [8] Fehring, R. J., Schneider, M., Raviele, K., Rodriguez, D., & Pruszynski, J. (2013). Randomized comparison of two Internet-supported fertility-awareness-based methods of family planning. Contraception, 88(1), 24–30. <https://doi.org/10.1016/j.contraception.2012.10.010>
- [9] Machine learning model for menstrual cycle phase classification and ovulation day detection based on sleeping heart rate under free-living conditions. ScienceDirect (2025). <https://www.sciencedirect.com/science/article/pii/S0010482525000551>
- [10] KOMUNTALE PAULA, NAKIRYA WHITNEY, KAREREIGANA COURTNEY (2025). Dataset for Menstrual Cycle Phase Prediction and Hygiene Guidance . IEEE Dataport. <https://dx.doi.org/10.21227/jf7f-s431>
- [11] Li, K., Urteaga, I., Shea, A., Vitzthum, V. J., Wiggins, C. H., & Elhadad, N. (2022). A predictive model for next cycle start date that accounts for adherence in menstrual self-tracking. Journal of the American Medical Informatics Association, 29(1), 3–11. <https://doi.org/10.1093/jamia/ocab182>
- [12] <https://www.kaggle.com/datasets/shreyasvedpathak/pcos-dataset>
- [13] Bhatta, A. (2024). Period Tracking Apps' Data Privacy Concerns: A Case Study Analysis. <https://anupamabhatta.github.io/docs/period-tracking-privacy.pdf>
- [14] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-Sampling Technique. Journal of Artificial Intelligence Research, 16, 321–357. <https://arxiv.org/abs/1106.1813>

- [15] Mary Sowjanya, A., & Mrudula, O. (2023). Effective treatment of imbalanced datasets in health care using modified SMOTE. *Applied Nanoscience*, 13(3), 1829–1840. <https://doi.org/10.1007/s13204-021-02063-4>
- [16] Global burden and projections of breast cancer incidence and mortality to 2050: GLOBOCAN data analysis. *Frontiers in Public Health* (2025). <https://pmc.ncbi.nlm.nih.gov/articles/PMC12611818/>
- [17] Yusuf, A., Iskandar, Y. H. P., Ab Hadi, I. S., Nasution, A., & Lean Keng, S. (2022). Breast awareness mobile apps for health education and promotion for breast cancer. *Frontiers in Public Health*, 10, 951641. <https://doi.org/10.3389/fpubh.2022.951641>
- [18] Feroz, A. S., De Vera, K., Bragagnolo, N. D., Saleem, S., Bhutta, Z., & Seto, E. (2022). Understanding the Needs of a Mobile Phone-Based Telemonitoring Program for Pregnant Women at High Risk for Pre-Eclampsia. *JMIR Formative Research*, 6(2), e32428. <https://formative.jmir.org/2022/2/e32428>
- [19] Mobilizing Breast Cancer Prevention Research Through Smartphone Apps: A Systematic Review. *PubMed Central* (2019). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6851054/>
- [20] Exploring Maternal and Infant Health App Development and Effectiveness Research: Scoping Review. *JMIR Pediatrics and Parenting* (2024). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10858421/>
- [21] Breast cancer statistics and resources. World Cancer Research Fund International / BCRF. <https://www.bcrf.org/breast-cancer-statistics-and-resources/>
- [22] <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>
- [23] Chowdhury, D., Das, A., Dey, A., et al. (2022). ABCanDroid: A Cloud Integrated Android App for Noninvasive Early Breast Cancer Detection Using Transfer Learning. *Sensors*, 22(3), 832. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8838592/>
- [24] Al-Dhabyani, W., Gomaa, M., Khaled, H., & Fahmy, A. (2020). Dataset of breast ultrasound images. *Data in Brief*, 28, 104863. (BUSI Dataset — 780 images, 600 women, National Cancer Institute Cairo) <https://doi.org/10.1016/j.dib.2019.104863>
- [25] Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>
- [26] Pawłowska, A., et al. (2024). Breast-Lesions-USG Dataset. The Cancer Imaging Archive. 256 biopsy-confirmed expert-annotated ultrasound images. <https://doi.org/10.7937/9WKK-Q141>